

WiFi Sense — Verification and Validation

1. Test environment

Item	Value
Room	5.0 × 4.0 m, sensor on the x=0 wall at y = 2.0 m
Node	ESP32 + HLK-LD2450 + HLK-LD2412 + AM312 PIR
Firmware	<code>firmware/sensor-node.yaml</code>
Server	<code>uvicorn server:app</code> , Python 3.11+
Client	Chrome / Firefox on PC; phone browser for FR7
Ground truth	tape measure, floor marked at 0.5 m intervals
Date of run	TODO

TODO — fill the date and note the firmware commit hash for each run, so a result can be traced back to a version of the code.

2. Acceptance criteria

The requirements need numbers before they can be tested. These are the thresholds I hold the product to; where PROJECT-INTRO is vaguer than this, this table is the stricter reading.

Req	Measurable criterion
FR1	A person entering the room is reported present within 3 s, in 10 of 10 entries
FR2	Reported position is within 0.5 m of the tape-measured position, at 8 marked points; frames arrive at ≥5 Hz
FR3	A seated, motionless person stays counted for ≥5 minutes

Req	Measurable criterion
FR4	The state text matches the number of dots on the map at all times
FR5	Zero dots appear for a person walking in the corridor behind the wall over a 5-minute run
FR6	No camera or microphone exists anywhere in the product (inspection, not a test)
FR7	Dashboard loads on a phone outside the home network within 10 s; notification arrives within 30 s of an armed detection
NFR1	Empty room, 8 hours: zero false presence events
NFR2	Dashboard latency (real movement → dot moves) under 1 s

3. Verification tests — does it work correctly?

No	Test	Instructions	Result
V1	Server starts and serves the dashboard	1. <code>uvicorn server:app</code> 2. open <code>http://localhost:8000</code> 3. page renders room grid	Works
V2	WebSocket connects and streams	1. open dashboard 2. read Status line 3. watch console frames	Works
V3	Simulator produces moving dots	1. start with radar offline 2. observe dots 3. check Source line	Work
V4	Node connects to the server	1. power node 2. start server 3. read Radar line	TODO — expect "sees no one" / "sees someone moving"
V5	Coordinate transform is correct	1. stand at marked point (2.0, 2.0) 2. read Position line	TODO — expect 2.0 ± 0.5 , 2.0 ± 0.5

No	Test	Instructions	Result
V6	Out-of-room points are rejected	1. hold a reflector beyond the wall 2. check no dot appears	TODO
V7	Hold timer in room centre	1. person stands mid-room 2. cover/stop detection 3. watch dot	TODO — expect yellow dot with counter, removed at 300 s
V8	Hold timer in door zone	1. walk into door zone 2. leave room	TODO — expect immediate removal, no yellow dot
V9	Hold timer in window zone	1. stand in window zone 2. stop being detected	TODO — expect yellow dot, removed at ~15 s
V10	Radar disconnect recovery	1. running live 2. unplug node 3. wait 30 s 4. replugin	TODO — expect "offline", fallback to simulator, reconnect within ~5 s
V11	Three targets tracked at once	1. three people walk the room 2. count dots	TODO
V12	Secrets are not in the repository	<code>git ls-files firmware/</code>	PASS — only <code>sensor-node.yaml</code> is tracked; <code>secrets.yaml</code> is ignored
V13	<code>/status</code> and <code>/api/state</code> return real data	1. <code>curl localhost:8000/api/state</code>	FAIL — returns the hard-coded placeholder <code>{"occupied": false, "people": 0, "source": "placeholder"}</code> . Expected: live state. Logged as R5 in REALISATION.md
V14	Live radar path runs end to end	1. start server with node online 2. walk the room	FAIL — blocked by R1–R4 (syntax and scope errors in <code>radar.py</code> ,

No	Test	Instructions	Result
			<code>server.py</code> , <code>dashboard.html</code>)

4. Validation tests — is it the right product?

No	Requirement	Validate	Result
T1	FR1 — detect that at least one person is present	1. start with empty room 2. walk in 3. read state text	TODO — expect "1 person" within 3 s
T2	FR2 — show each person's position on the map, $\geq 5 \times/s$	1. stand on each of 8 floor marks 2. compare dot to tape measure 3. record error per point	TODO — expect ≤ 0.5 m error at every point
T3	FR3 — a motionless person stays counted	1. sit still in a chair 2. do not move for 6 minutes 3. watch dot and counter	TODO — expect the person never disappears
T4	FR4 — current state shown in text	1. vary 0, 1, 2, 3 people 2. compare text against dots each time	TODO
T5	FR5 — ignore movement outside the room	1. have someone walk the corridor behind the wall for 5 min 2. count dots	TODO — expect zero
T6	FR6 — no camera or microphone	1. inspect the hardware 2. inspect the code for any image/audio handling	PASS — the product contains a radar, a PIR and a microcontroller; no camera or microphone exists to be compromised
T7	FR7 — viewable from the phone off-	1. leave the house 2. open the dashboard over Tailscale 3. have	NOT RUN — Tailscale not configured and notifications not implemented

No	Requirement	Validate	Result
	network + notification	someone enter the armed room	
T8	NFR1 — no false presence in an empty room	1. leave the house for 8 h with the server logging 2. review for any presence event	TODO — the fan and curtains are the suspects here
T9	NFR2 — the map keeps up with reality	1. walk while a second person watches the screen 2. estimate lag	TODO — expect under 1 s
T10	Usable by someone who did not build it	1. hand a housemate the dashboard 2. ask "is anyone in my room, and where?" 3. observe without explaining	NOT RUN

5. Test log

Each run gets a row. Keep failures, do not overwrite them — a fixed failure is better evidence than a test that was always green.

Date	Test	Result	What I changed
2026-09-23	V12	PASS	—
2026-09-23	V13	FAIL	logged as R5, not fixed yet
2026-09-23	V14	FAIL	logged as R1–R4, fixing next

6. Evidence

Evidence	File	Covers
First live-hardware demo recording	docs/first_real_life_demotest.mp4	V4, V5 (informally)

Evidence	File	Covers
Dashboard screenshots, simulator	TODO	V3
Dashboard screenshots, live radar	TODO	T1, T2
Accuracy measurement photos or table	TODO	T2
Overnight empty-room log	TODO	T8
Stakeholder / peer feedback notes	TODO	T10

7. Conclusion

TODO — write after the tests have actually run. State per requirement whether it is met, which ones failed and what changed as a result, and what the product cannot yet do. An honest "FR7 is not met, here is why" is worth more here than a conclusion that quietly skips it.

What can already be said: the simulator path is a complete product end to end, and the live radar path is one debugging session away from testable. The requirements that are certainly not met yet are FR7 (remote access and notifications) and the multi-person half of FR3.